Colin Sauder

303-808-2458 ● Evergreen CO ● cfsauder@gmail.com ● https://tactical-ostrich.github.io/personal-website

Education

Colorado School of Mines

3.74 GPA

- Bachelors of Science in Engineering Physics — August 2024 - May 2026
- Masters of Science in Applied Physics — May 2026 - December 2027
- Masters of Science in Mechanical Engineering — May 2026 - December 2027

Relevant coursework:

Circuits, Digital Circuits, Intermediate Mechanics, Mathematical Physics, Advanced Electrodynamics, Advanced Lab 2, Field Session

University of Denver — September 2022 - May 2024 (Transferred)

3.87 Final GPA

- Bachelors of Science in Physics (Transferred to Colorado School of Mines)
- Minor in Mathematics

Relevant Coursework:

Statics, Programing (Python), Astrophysics: Radiative Process, Error and Uncertainty Analysis, Modern Physics 1 & 2, Modern Physics Lab

Projects / Coursework

Van Der Graff Generator

- Built a working, low voltage, Van Der Graff generator independently from a vinyl / nylon triboelectric effect.
- Used it to demonstrate the potential for a lint roller using electrostatic force.

Jewelrymaking

- Created several different jewelry pieces (rings, pendants, bracelets) through casting and soldering.
- Independently learned to use various tools and methods to create and refine these different jewelry pieces.

Field Session

- Created a receiver for a spark gap transmitter using physical circuitry and labview to output a morse code readout using the length of the spark produced.
- Worked in a small team to create a 3d projection using both linearly and circularly polarized laser light.
- Learned and experimented with different vacuum systems to understand their operation and how to utilize them for different functions like deposition or mass spectrometry.

Refining a Vacuum system for eventual use for DC Sputtering

- Serviced and installed components on a vacuum chamber continuing previous students' works to create a vacuum chamber for DC sputtering.
- Designed a latching system to ensure proper sealing for the initial pump down of the chamber.
- Designed and plumbed the water cooling system to back the turbo and DC sputter head.

Advanced Lab

- Re-created historical physics experiments in optics and other fields (Michelson-Moorley, Young's double slit, etc).
- Detected different types of radiation and fundamental particles from background radiation.

Work Experience

Colorado School of Mines, Golden, CO — August 2025 - December 2025

Physics 2 Studio TA

- Reinforced foundational electromagnetic concepts for students during studio.
- Tutored students to help with homework problems.

Bergen Bark Inn, Evergreen, CO — May 2024 - September 2025

Kennel Tech / Pet Pro

- Cleaned dog kennels and ensured the wellbeing and health of the up to 50 animals in our care.
- Worked independently and in small teams (2-4) to close the kennel.

Community Involvement

Denver Museum of Nature and Science — March 2025 - Present

Facilitator volunteer

- Helped facilitate childhood and community learning in various scientific fields using museum carts.
- Communicated and taught foundational concepts of experimentation and robotics to children through hands-on learning.

Skills

Technical

- Microsoft Excel, Word, Python, C++, SolidWorks, Arduino, Wolfram Mathematica, HTML/CSS, LaTeX

Laboratory

- Labview, Data Collection, Data Analysis